START

7th International Conference on Bacteriophage Research and Antimicrobial Resistance

Lytic Legends Edition · 13 rounds of plaques, probes and punchlines

Teams

Lytic Legends

Dr. Ruma

Ms. Priya

Mr. Karan

Biofilm Breakers

Dr. Rupali

Dr. Prachi

Mr. Prajasattak

Phage Pharmers

Team member 1

Team member 2

Team member 3

Virion Architects

Team member 1

Team member 2

Team member 3

Rounds

1. GENERAL QUESTIONS
2. HISTORY OF BACTERIOPHAGES
3. PHAGE RELATED PROTOCOLS
4. PHAGE INFORMATICS
5. PHAGE IDENTIFICATION
6. IDENTIFY THE FIGURE
7. PHAGE THERAPY & CLINICAL FRONTIERS NEW
8. PHAGE ACRONYMS
9. PHAGE ECOLOGY & ONE HEALTH NEW
10. BACTERIAL DEFENSES & PHAGE COUNTER-DE

BREAK (5 min)

11. RAPID FIRE
12. TIE-BREAKER ROUND
13. LYTIC LAUGHS

Markings

Rounds 1-5	+10 / 0	30 sec
Rounds 6-10	+10 / -5	30 sec
Rapid Fire	+5 / 0, no passing	60 sec / team
Tie-Breaker	+20 / -10, passed +5	60 sec
Lytic Laughs	+10 / 0	30 sec

Each MCQ round: 8 team questions + 2 audience questions

1

GENERAL QUESTIONS

Correct +10 | Wrong 0 | 30 sec

ROUND 1

Bacteriophages are best described as

A Viruses that specifically infect and replicate within bacteria

B Toxic proteins secreted by bacteria

C Bacteria that prey on other bacteria

D Viruses that infect human cells

ROUND 1

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A Viruses that specifically infect and replicate within bacteria

B Toxic proteins secreted by bacteria

C Bacteria that prey on other bacteria

D Viruses that infect human cells

ROUND 1

The estimated number of phage particles on Earth is approximately

A 10^15 (a quadrillion)

B 10^21 (a sextillion)

C 10^31 (ten nonillion)

D 10^45

ROUND 1

The estimated number of phage particles on Earth is approximately

A 10^15 (a quadrillion)

B 10^21 (a sextillion)

C 10^31 (ten nonillion)

D 10^45

ROUND 1

Which structure is NOT part of a typical tailed phage virion?

A Tail fibers

B Baseplate

C Ribosomes

D Icosahedral capsid

ROUND 1

Which structure is NOT part of a typical tailed phage virion?

A Tail fibers

B Baseplate

C Ribosomes

D Icosahedral capsid

ROUND 1

The phage component that primarily determines host specificity is the

A Genome terminus

B Portal vertex

C Capsid interior

D Receptor-binding protein / tail fiber

ROUND 1

The phage component that primarily determines host specificity is the

A Genome terminus

B Portal vertex

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ROUND 1

A phage that can ONLY replicate through the lytic cycle is termed

A Virulent

B Temperate

C Defective

D Filamentous

ROUND 1

A phage that can ONLY replicate through the lytic cycle is termed

A Virulent

B Temperate

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ROUND 1

A clear zone formed on a confluent bacterial lawn is called a

A Plaque

B Biofilm patch

C Colony

D Halo

ROUND 1

A clear zone formed on a confluent bacterial lawn is called a

A Plaque

B Biofilm patch

C Colony

D Halo

ROUND 1

The typical diameter of a tailed phage capsid falls in the range

A 500·900 nm

B 1·5 µm

C 5·10 µm

D 20·200 nm (up to ~300 nm for jumbos)

ROUND 1

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A 500·900 nm

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D 20·200 nm (up to ~300 nm for jumbos)

ROUND 1

Which statement about phages is TRUE?

A They can grow on cell-free agar

B They are obligate intracellular parasites of bacteria

C They always kill their host

D They carry their own ribosomes for translation

ROUND 1

Which statement about phages is TRUE?

A They can grow on cell-free agar

B They are obligate intracellular parasites of bacteria

C They always kill their host

D They carry their own ribosomes for translation

ROUND 1

AUDIENCE

Which is generally SMALLER?

A A red blood cell

B A bacteriophage

C A yeast cell

D A bacterium

ROUND 1

AUDIENCE

Which is generally SMALLER?

A A red blood cell

B A bacteriophage

C A yeast cell

D A bacterium

ROUND 1

AUDIENCE

Phage contamination of starter cultures is a classic industrial problem in which fermentation?

A Bread baking

B Cheese and yogurt making

C Beer brewing

D Wine making

ROUND 1

AUDIENCE

Phage contamination of starter cultures is a classic industrial problem in which fermentation?

A Bread baking

B Cheese and yogurt making

C Beer brewing

D Wine making

2

HISTORY OF BACTERIOPHAGES

Correct +10 | Wrong 0 | 30 sec

ROUND 2

In which year did Félix d'Hérelle coin the term ·bacteriophage·?

A 1926

B 1896

C 1917

D 1915

ROUND 2

In which year did Félix d'Hérelle coin the term ·bacteriophage·?

A 1926

B 1896

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D 1915

ROUND 2

Ernest Hankin (1896) observed antibacterial activity against Vibrio cholerae in the waters of which rivers?

A The Ganges and the Yamuna

B The Nile and the Euphrates

C The Thames and the Seine

D The Amazon and the Orinoco

ROUND 2

Ernest Hankin (1896) observed antibacterial activity against Vibrio cholerae in the waters of which rivers?

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ROUND 2

The 1943 fluctuation test, demonstrating that mutations arise spontaneously, used resistance to phage and was performed by

A Ellis and Delbrück

B Luria and Delbrück

C Hershey and Chase

D Zinder and Lederberg

ROUND 2

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ROUND 2

The Hershey·Chase experiment (1952) labelled phage components with which isotopes?

A 14C and 3H

B 32P and 35S

C 35S and 14C for DNA

D 15N and 13C

ROUND 2

The Hershey·Chase experiment (1952) labelled phage components with which isotopes?

A 14C and 3H

B 32P and 35S

C 35S and 14C for DNA

D 15N and 13C

ROUND 2

The first complete genome ever sequenced (Sanger, 1977) belonged to which phage?

A Phage lambda

B Phage T4

C Phage ·X174

D Phage M13

ROUND 2

The first complete genome ever sequenced (Sanger, 1977) belonged to which phage?

A Phage lambda

B Phage T4

C Phage ·X174

D Phage M13

ROUND 2

The Eliava Institute, a historic centre of phage therapy, is located in

A Tbilisi, Georgia

B Lyon, France

C Moscow, Russia

D Warsaw, Poland

ROUND 2

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ROUND 2

Generalized transduction was discovered by Zinder and Lederberg (1952) using which phage·host system?

A Phage P22 in Salmonella

B Phage lambda in E. coli

C Phage T2 in E. coli

D Phage P1 in E. coli

ROUND 2

Generalized transduction was discovered by Zinder and Lederberg (1952) using which phage·host system?

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ROUND 2

Western interest in phage therapy collapsed after the 1940s mainly because of

A Loss of phage stocks in WWII

B A ban on virus research

C Patent disputes

D The discovery of antibiotics

ROUND 2

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ROUND 2

AUDIENCE

The 2017 compassionate-use case that brought phage therapy to global headlines treated a patient with

A Candida auris sepsis

B XDR tuberculosis

C MRSA endocarditis

D Disseminated Acinetobacter baumannii infection

ROUND 2

AUDIENCE

The 2017 compassionate-use case that brought phage therapy to global headlines treated a patient with

A Candida auris sepsis

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ROUND 2

AUDIENCE

The 1969 Nobel Prize (Delbrück, Hershey, Luria) honoured work on the replication and genetic structure of

A Influenza virus

B Bacteriophages

C Poliovirus

D Retroviruses

ROUND 2

AUDIENCE

The 1969 Nobel Prize (Delbrück, Hershey, Luria) honoured work on the replication and genetic structure of

A Influenza virus

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3

PHAGE RELATED PROTOCOLS

Correct +10 | Wrong 0 | 30 sec

ROUND 3

The double agar overlay method is primarily used for

A Plaque assay and phage enumeration

B Antibiotic susceptibility testing

C Genome assembly

D Protein purification

ROUND 3

The double agar overlay method is primarily used for

A Plaque assay and phage enumeration

B Antibiotic susceptibility testing

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D Protein purification

ROUND 3

The soft (top) agar used in plaque assays typically has an agar concentration of

A 3·4%

B 1.5·2.0%

C 0.3·0.7%

D 8·10%

ROUND 3

The soft (top) agar used in plaque assays typically has an agar concentration of

A 3·4%

B 1.5·2.0%

C 0.3·0.7%

D 8·10%

ROUND 3

Ca2+ and Mg2+ are added to phage assay media mainly because

A Many phages need divalent cations for efficient adsorption

B They raise the agar melting point

C They prevent fungal contamination

D They stain the plaques

ROUND 3

Ca2+ and Mg2+ are added to phage assay media mainly because

A Many phages need divalent cations for efficient adsorption

B They raise the agar melting point

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ROUND 3

Which method counts ONLY infectious phage particles?

A Flow cytometry with SYBR Gold

B qPCR of phage DNA

C TEM particle counts

D Plaque assay

ROUND 3

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ROUND 3

To enrich phages from an environmental sample, the first step is usually to

A Autoclave the sample

B Co-culture it with a log-phase host bacterium

C Freeze it at −80 °C for a week

D Add chloroform immediately

ROUND 3

To enrich phages from an environmental sample, the first step is usually to

A Autoclave the sample

B Co-culture it with a log-phase host bacterium

C Freeze it at −80 °C for a week

D Add chloroform immediately

ROUND 3

Chloroform is added to a phage lysate to

A Precipitate the phages

B Stain phage DNA

C Kill residual bacteria and prevent bacterial growth

D Lower the pH

ROUND 3

Chloroform is added to a phage lysate to

A Precipitate the phages

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C Kill residual bacteria and prevent bacterial growth

D Lower the pH

ROUND 3

Which of the following is NOT a phage concentration method?

A PEG precipitation

B Southern blotting

C Tangential flow filtration

D CsCl density-gradient ultracentrifugation

ROUND 3

Which of the following is NOT a phage concentration method?

A PEG precipitation

B Southern blotting

C Tangential flow filtration

D CsCl density-gradient ultracentrifugation

ROUND 3

Efficiency of plating (EOP) compares plaque production on a test strain relative to

A The adsorption rate

B The primary host strain

C The burst size

D The qPCR genome count

ROUND 3

Efficiency of plating (EOP) compares plaque production on a test strain relative to

A The adsorption rate

B The primary host strain

C The burst size

D The qPCR genome count

ROUND 3

AUDIENCE

In a spot test, a clear zone where phage suspension was dropped on a bacterial lawn indicates

A The strain is sensitive (lysis)

B A mutation arose

C Medium contamination

D The strain is resistant

ROUND 3

AUDIENCE

In a spot test, a clear zone where phage suspension was dropped on a bacterial lawn indicates

A The strain is sensitive (lysis)

B A mutation arose

C Medium contamination

D The strain is resistant

ROUND 3

AUDIENCE

In a plaque assay, each plaque originates from

A A single bacterial cell

B A free DNA fragment

C An antibiotic disc

D A single infectious phage particle

ROUND 3

AUDIENCE

In a plaque assay, each plaque originates from

A A single bacterial cell

B A free DNA fragment

C An antibiotic disc

D A single infectious phage particle

4

PHAGE INFORMATICS

Correct +10 | Wrong 0 | 30 sec

ROUND 4

Which tool is dedicated to predicting prophage regions inside bacterial genomes?

A PHASTER

B tRNAscan-SE

C Clustal Omega

D SignalP

ROUND 4

Which tool is dedicated to predicting prophage regions inside bacterial genomes?

A PHASTER

B tRNAscan-SE

C Clustal Omega

D SignalP

ROUND 4

CheckV is used to

A Detect AMR genes in bacteria

B Design PCR primers

C Count plaques automatically

D Estimate completeness and quality of viral metagenome-assembled

ROUND 4

CheckV is used to

A Detect AMR genes in bacteria

B Design PCR primers

C Count plaques automatically

D Estimate completeness and quality of viral metagenome-assembled

ROUND 4

VIRIDIC calculates intergenomic similarity to demarcate viral

A Promoters

B tRNAs

C Codon usage

D Species and genera

ROUND 4

VIRIDIC calculates intergenomic similarity to demarcate viral

A Promoters

B tRNAs

C Codon usage

D Species and genera

ROUND 4

vConTACT2 clusters viral genomes using

A Gene-sharing networks

B k-mer GC content

C 16S rRNA alignment

D CRISPR spacers only

ROUND 4

vConTACT2 clusters viral genomes using

A Gene-sharing networks

B k-mer GC content

C 16S rRNA alignment

D CRISPR spacers only

ROUND 4

PHROGs are

A Phage orthologous groups used for functional annotation

B Plaque morphology classes

C Phage restriction enzymes

D A phage growth medium

ROUND 4

PHROGs are

A Phage orthologous groups used for functional annotation

B Plaque morphology classes

C Phage restriction enzymes

D A phage growth medium

ROUND 4

crAssphage was originally discovered in 2014 by

A Plaque assay on E. coli

B Sequencing of seawater viromes

C Computational re-analysis of human gut metagenomes

D TEM of sewage

ROUND 4

crAssphage was originally discovered in 2014 by

A Plaque assay on E. coli

B Sequencing of seawater viromes

C Computational re-analysis of human gut metagenomes

D TEM of sewage

ROUND 4

Which approach can predict the host of a novel phage from sequence data?

A MALDI-TOF of the capsid

B CRISPR spacer and prophage homology matching

C Gram staining of the virome

D PFGE

ROUND 4

Which approach can predict the host of a novel phage from sequence data?

A MALDI-TOF of the capsid

B CRISPR spacer and prophage homology matching

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D PFGE

ROUND 4

PhageTerm is used to identify

A Optimal cocktail composition

B Genome termini and the packaging mechanism (cos, pac, headful)

C Latent period

D Phage titer

ROUND 4

PhageTerm is used to identify

A Optimal cocktail composition

B Genome termini and the packaging mechanism (cos, pac, headful)

C Latent period

D Phage titer

ROUND 4

AUDIENCE

The complete collection of viruses in a sample is called the

A Plasmidome

B Proteome

C Virome

D Microbiome

ROUND 4

AUDIENCE

The complete collection of viruses in a sample is called the

A Plasmidome

B Proteome

C Virome

D Microbiome

ROUND 4

AUDIENCE

A prophage region whose GC content differs strongly from the rest of the bacterial chromosome suggests

A A smaller capsid

B Recently acquired foreign DNA

C An RNA genome

D A better plaque morphology

ROUND 4

AUDIENCE

A prophage region whose GC content differs strongly from the rest of the bacterial chromosome suggests

A A smaller capsid

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5

PHAGE IDENTIFICATION

Correct +10 | Wrong 0 | 30 sec

ROUND 5

A phage with an icosahedral head and a long CONTRACTILE tail shows which morphotype?

A Siphovirus

B Inovirus

C Podovirus

D Myovirus

ROUND 5

A phage with an icosahedral head and a long CONTRACTILE tail shows which morphotype?

A Siphovirus

B Inovirus

C Podovirus

D Myovirus

ROUND 5

Filamentous phages such as M13, which are secreted without lysing the host, belong to which group?

A Tectiviruses

B Corticoviruses

C Myoviruses

D Inoviruses (filamentous)

ROUND 5

Filamentous phages such as M13, which are secreted without lysing the host, belong to which group?

A Tectiviruses

B Corticoviruses

C Myoviruses

D Inoviruses (filamentous)

ROUND 5

Which phage is an ENVELOPED virus with a double-stranded RNA genome?

A Phage lambda

B Phage ·6

C Phage T4

D Phage MS2

ROUND 5

Which phage is an ENVELOPED virus with a double-stranded RNA genome?

A Phage lambda

B Phage ·6

C Phage T4

D Phage MS2

ROUND 5

The genome of the levivirus MS2 is

A ssRNA(+)

B ssDNA

C dsRNA

D dsDNA

ROUND 5

The genome of the levivirus MS2 is

A ssRNA(+)

B ssDNA

C dsRNA

D dsDNA

ROUND 5

In siphoviruses, reversible host recognition is mediated mainly by

A Collar whiskers

B Distal tail fibers / tail-tip proteins

C The portal vertex

D The internal core

ROUND 5

In siphoviruses, reversible host recognition is mediated mainly by

A Collar whiskers

B Distal tail fibers / tail-tip proteins

C The portal vertex

D The internal core

ROUND 5

·Jumbo· phages are defined by

A Tails longer than 500 nm

B Genomes larger than ~200 kb

C RNA genomes

D Titers above 10^12 PFU/mL

ROUND 5

·Jumbo· phages are defined by

A Tails longer than 500 nm

B Genomes larger than ~200 kb

C RNA genomes

D Titers above 10^12 PFU/mL

ROUND 5

Which phage-encoded tubulin homolog positions the capsids of jumbo phage ϕKZ inside the host?

A MreB

B FtsZ

C PhuZ

D ParM

ROUND 5

Which phage-encoded tubulin homolog positions the capsids of jumbo phage ϕKZ inside the host?

A MreB

B FtsZ

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ROUND 5

During T4 infection, the structure that CONTRACTS to drive the tail tube across the envelope is the

A Tail sheath

B Whiskers

C Portal

D Collar

ROUND 5

During T4 infection, the structure that CONTRACTS to drive the tail tube across the envelope is the

A Tail sheath

B Whiskers

C Portal

D Collar

ROUND 5

AUDIENCE

Under TEM, a tailed phage with a small head and a very SHORT non-contractile tail is a

A Podovirus

B Myovirus

C Siphovirus

D Filamentous phage

ROUND 5

AUDIENCE

Under TEM, a tailed phage with a small head and a very SHORT non-contractile tail is a

A Podovirus

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C Siphovirus

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ROUND 5

AUDIENCE

Turbid (cloudy) plaques usually suggest

A An RNA phage

B Medium contamination

C A temperate phage forming lysogens

D A strictly lytic phage

ROUND 5

AUDIENCE

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A An RNA phage

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6

IDENTIFY THE FIGURE

Correct +10 | Wrong -5 | 30 sec

ROUND 6

In this one-step growth curve, the arrowed flat phase BEFORE the rise represents the

A Burst phase

B Latent (eclipse) period

C Lysis from without

D Adsorption phase

ROUND 6

In this one-step growth curve, the arrowed flat phase BEFORE the rise represents the

A Burst phase

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C Lysis from without

D Adsorption phase

ROUND 6

The clear HALO extending beyond the plaque edge in this plate is most likely due to

A Phage-resistant mutants

B Lysis from without

C Extracellular depolymerase degrading the capsule/EPS

D An antibiotic in the medium

ROUND 6

The clear HALO extending beyond the plaque edge in this plate is most likely due to

A Phage-resistant mutants

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D An antibiotic in the medium

ROUND 6

The arrowed structure of this virion, responsible for reversible host recognition, is the

A Contractile sheath

B Long tail fiber

C Collar

D Baseplate

ROUND 6

The arrowed structure of this virion, responsible for reversible host recognition, is the

A Contractile sheath

B Long tail fiber

C Collar

D Baseplate

ROUND 6

The infection state depicted in this diagram is

A Lysogeny (prophage integrated in the chromosome)

B Lysis from without

C Chronic filamentous infection

D Pseudolysogeny

ROUND 6

The infection state depicted in this diagram is

A Lysogeny (prophage integrated in the chromosome)

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D Pseudolysogeny

ROUND 6

The technique depicted here is the

A Spread-plate CFU count

B Streak-plate isolation

C Disc diffusion assay

D Double agar overlay plaque assay

ROUND 6

The technique depicted here is the

A Spread-plate CFU count

B Streak-plate isolation

C Disc diffusion assay

D Double agar overlay plaque assay

ROUND 6

After ultracentrifugation, the visible band in the middle of this CsCl tube contains

A Purified phage particles at their buoyant density

B Lipopolysaccharide

C Host chromosomal DNA

D Ribosomal proteins

ROUND 6

After ultracentrifugation, the visible band in the middle of this CsCl tube contains

A Purified phage particles at their buoyant density

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ROUND 6

The MIDDLE virion of this trio (long, flexible, non-contractile tail) is a

A Siphovirus morphotype

B Myovirus morphotype

C Podovirus morphotype

D Filamentous inovirus

ROUND 6

The MIDDLE virion of this trio (long, flexible, non-contractile tail) is a

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ROUND 6

To penetrate the matrix shown in this biofilm diagram, the most useful phage-encoded enzymes are

A DNA ligases

B Integrases

C RNA polymerases

D Depolymerases

ROUND 6

To penetrate the matrix shown in this biofilm diagram, the most useful phage-encoded enzymes are

A DNA ligases

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C RNA polymerases

D Depolymerases

ROUND 6

The assay shown on this plate, used for quick host-range screening, is the

A Coagulase test

B Disk diffusion test

C Spot test (spot assay)

D CAMP test

ROUND 6

The assay shown on this plate, used for quick host-range screening, is the

A Coagulase test

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ROUND 6

Plate A shows clear plaques, plate B turbid plaques. Which plate most likely contains a TEMPERATE phage?

A Both plates

B Plate B

C Neither plate

D Plate A

ROUND 6

Plate A shows clear plaques, plate B turbid plaques. Which plate most likely contains a TEMPERATE phage?

A Both plates

B Plate B

C Neither plate

D Plate A

7

PHAGE THERAPY & CLINICAL FRONTIERS

Correct +10 | Wrong -5 | 30 sec

ROUND 7

NEW TOPIC

The 2019 Nature Medicine report of an engineered three-phage cocktail successfully treated which infection in a cystic fibrosis patient?

A Staphylococcus aureus osteomyelitis

B Disseminated Mycobacterium abscessus infection

C Pseudomonas aeruginosa pneumonia

D Acinetobacter baumannii sepsis

ROUND 7

NEW TOPIC

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ROUND 7

NEW TOPIC

Since 2018, Belgium delivers therapeutic phages to patients as

A Over-the-counter medicines

B Vaccines

C Medical devices

D Active pharmaceutical ingredients within magistral preparations

ROUND 7

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ROUND 7

NEW TOPIC

The PhagoBurn trial (Lancet ID, 2018) evaluated phage therapy against which pathogen in burn wounds?

A Escherichia coli

B Staphylococcus aureus

C Pseudomonas aeruginosa

D Klebsiella pneumoniae

ROUND 7

NEW TOPIC

The PhagoBurn trial (Lancet ID, 2018) evaluated phage therapy against which pathogen in burn wounds?

A Escherichia coli

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ROUND 7

NEW TOPIC

LBP-EC01, the first engineered (CRISPR-Cas3-enhanced) phage tested in a randomized placebo-controlled trial, targets

A Escherichia coli in urinary tract infection

B Clostridioides difficile colitis

C Pseudomonas aeruginosa in CF

D Staphylococcus aureus bacteremia

ROUND 7

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ROUND 7

NEW TOPIC

·Phage steering· mitigates AMR by

A Increasing the MOI until resistance collapses

B Selecting phage-resistant mutants whose fitness cost re-sensitizes them to antibiotics

C Blocking phage receptors with antibodies

D Using phages to inactivate antibiotics

ROUND 7

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A Increasing the MOI until resistance collapses

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ROUND 7

NEW TOPIC

A key immunological limitation of repeated intravenous phage dosing is

A Permanent T-cell exhaustion

B Neutralizing antibodies that accelerate phage clearance

C Phages cannot enter the bloodstream

D Immediate IgE anaphylaxis in every patient

ROUND 7

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ROUND 7

NEW TOPIC

In the United States, compassionate-use phage therapy is typically authorized through

A An FDA emergency/expanded-access IND (eIND)

B EPA registration

C An OTC monograph

D A USDA rule

ROUND 7

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D A USDA rule

ROUND 7

NEW TOPIC

Before clinical use, a therapeutic phage genome must be screened to confirm

A Absence of toxin, AMR and virulence genes

B GC content above 60%

C Uracil instead of thymine

D Presence of an integrase gene

ROUND 7

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A Absence of toxin, AMR and virulence genes

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ROUND 7

NEW TOPIC

AUDIENCE

Commercial phage preparations sprayed on ready-to-eat food to control Listeria are an example of

A Pasteurization

B Vaccination

C Fermentation

D Biocontrol

ROUND 7

NEW TOPIC

AUDIENCE

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A Pasteurization

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ROUND 7

NEW TOPIC

AUDIENCE

Compared with antibiotics, a distinctive advantage of phages is that they

A Kill human cells selectively

B Never encounter resistance

C Self-amplify at the site of infection

D Work without any bacteria present

ROUND 7

NEW TOPIC

AUDIENCE

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A Kill human cells selectively

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8

PHAGE ACRONYMS

Correct +10

Wrong -5

30 sec

ROUND 8

In phage work, EOP stands for

A Endotoxin elimination process

B Efficiency of plating

C Effective optical power

D Enzyme oligomerization protocol

ROUND 8

In phage work, EOP stands for

A Endotoxin elimination process

B Efficiency of plating

C Effective optical power

D Enzyme oligomerization protocol

ROUND 8

In phage genomics, AMGs are

A Auxiliary metabolic genes

B Adaptive mutation groups

C Amplified mobile genes

D Ancestral marker genes

ROUND 8

In phage genomics, AMGs are

A Auxiliary metabolic genes

B Adaptive mutation groups

C Amplified mobile genes

D Ancestral marker genes

ROUND 8

Acr proteins, encoded by many phages, are

A Acidic capsid repeats

B Active replication complexes

C Anti-CRISPR proteins

D Adsorption cofactor receptors

ROUND 8

Acr proteins, encoded by many phages, are

A Acidic capsid repeats

B Active replication complexes

C Anti-CRISPR proteins

D Adsorption cofactor receptors

ROUND 8

In phage biology, RBPs are

A Receptor-binding proteins

B Repeat-binding polymers

C Restriction barrier proteins

D Ribosome-binding pockets

ROUND 8

In phage biology, RBPs are

A Receptor-binding proteins

B Repeat-binding polymers

C Restriction barrier proteins

D Ribosome-binding pockets

ROUND 8

In phage·mucus research, the BAM model stands for

A Bacteriophage adherence to mucus

B Bacterial adhesion matrix

C Biofilm-associated microcolony

D Buffered agar medium

ROUND 8

In phage·mucus research, the BAM model stands for

A Bacteriophage adherence to mucus

B Bacterial adhesion matrix

C Biofilm-associated microcolony

D Buffered agar medium

ROUND 8

ICTV, the body that ratifies virus taxonomy, stands for

A International Council for Therapeutic Viruses

B Integrated Catalogue of Tested Viruses

C Institute of Clinical Trials in Virology

D International Committee on Taxonomy of Viruses

ROUND 8

ICTV, the body that ratifies virus taxonomy, stands for

A International Council for Therapeutic Viruses

B Integrated Catalogue of Tested Viruses

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ROUND 8

In phage formulation, LAL assay reagent is derived from

A Lysed adenovirus libraries

B Limulus (horseshoe crab) amebocyte lysate

C Lactic acid bacteria

D Lipid antigen liposomes

ROUND 8

In phage formulation, LAL assay reagent is derived from

A Lysed adenovirus libraries

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ROUND 8

The phage cytoskeletal protein PhuZ belongs to which protein family?

A Keratin

B Myosin

C Actin

D Tubulin

ROUND 8

The phage cytoskeletal protein PhuZ belongs to which protein family?

A Keratin

B Myosin

C Actin

D Tubulin

ROUND 8

AUDIENCE

In One-Health discussions, AMR stands for

A Adaptive mutation rate

B Antibiotic modifier region

C Antimicrobial resistance

D Acquired mobile resistance

ROUND 8

AUDIENCE

In One-Health discussions, AMR stands for

A Adaptive mutation rate

B Antibiotic modifier region

C Antimicrobial resistance

D Acquired mobile resistance

ROUND 8

AUDIENCE

In the phage morphology lab, TEM stands for

A Tunneling electron mapping

B Transmission electron microscopy

C Total evanescent measurement

D Thermal emission microscopy

ROUND 8

AUDIENCE

In the phage morphology lab, TEM stands for

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9

PHAGE ECOLOGY & ONE HEALTH

Correct +10 | Wrong -5 | 30 sec

ROUND 9

NEW TOPIC

The ·viral shunt· refers to

A Phage lysis redirecting microbial biomass into dissolved organic matter

B Aerosolization of virions

C Viral transport in the bloodstream

D Phage migration through soil

ROUND 9

NEW TOPIC

The ·viral shunt· refers to

A Phage lysis redirecting microbial biomass into dissolved organic matter

B Aerosolization of virions

C Viral transport in the bloodstream

D Phage migration through soil

ROUND 9

NEW TOPIC

A millilitre of surface seawater typically contains about

A 10^3 phages

B 10^12 phages

C 10^7 phages

D 10^15 phages

ROUND 9

NEW TOPIC

A millilitre of surface seawater typically contains about

A 10^3 phages

B 10^12 phages

C 10^7 phages

D 10^15 phages

ROUND 9

NEW TOPIC

Cyanophages infecting Prochlorococcus often carry which host-derived gene to sustain photosynthesis during infection?

A nifH

B psbA (photosystem II D1 protein)

C rpoB

D atpD

ROUND 9

NEW TOPIC

Cyanophages infecting Prochlorococcus often carry which host-derived gene to sustain photosynthesis during infection?

A nifH

B psbA (photosystem II D1 protein)

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ROUND 9

NEW TOPIC

·Kill-the-winner· dynamics predicts that

A Rare bacterial species are lysed first

B The most abundant host is preferentially hit by lytic phages

C Phages prefer dormant hosts

D Lysogeny dominates in rich environments

ROUND 9

NEW TOPIC

·Kill-the-winner· dynamics predicts that

A Rare bacterial species are lysed first

B The most abundant host is preferentially hit by lytic phages

C Phages prefer dormant hosts

D Lysogeny dominates in rich environments

ROUND 9

NEW TOPIC

Which phage is among the most prevalent human-associated viruses on Earth, carried in the gut of a large fraction of adults?

A Phage ·6

B Phage lambda

C crAssphage

D Phage T4

ROUND 9

NEW TOPIC

Which phage is among the most prevalent human-associated viruses on Earth, carried in the gut of a large fraction of adults?

A Phage ·6

B Phage lambda

C crAssphage

D Phage T4

ROUND 9

NEW TOPIC

The cholera toxin genes (ctxAB) are encoded by

A Phage P1

B Phage lambda

C Phage Mu

D Phage CTX·

ROUND 9

NEW TOPIC

The cholera toxin genes (ctxAB) are encoded by

A Phage P1

B Phage lambda

C Phage Mu

D Phage CTX·

ROUND 9

NEW TOPIC

Diphtheria toxin is encoded by

A Phage CTX·

B The ·-prophage (corynephage ·)

C Phage P22

D Phage ·KZ

ROUND 9

NEW TOPIC

Diphtheria toxin is encoded by

A Phage CTX·

B The ·-prophage (corynephage ·)

C Phage P22

D Phage ·KZ

ROUND 9

NEW TOPIC

Shiga toxin genes (stx) carried by lambdoid prophages are central to the virulence of

A Shiga-toxigenic E. coli (STEC)

B Salmonella Typhi

C Listeria monocytogenes

D Campylobacter jejuni

ROUND 9

NEW TOPIC

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A Shiga-toxigenic E. coli (STEC)

B Salmonella Typhi

C Listeria monocytogenes

D Campylobacter jejuni

ROUND 9

NEW TOPIC

AUDIENCE

Municipal sewage is extremely rich in phages mainly because

A High bacterial densities support massive phage replication

B Phages resist freezing

C UV light creates virions

D Chlorine selects for viruses

ROUND 9

NEW TOPIC

AUDIENCE

Municipal sewage is extremely rich in phages mainly because

A High bacterial densities support massive phage replication

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D Chlorine selects for viruses

ROUND 9

NEW TOPIC

AUDIENCE

Phages are part of ·One Health· thinking because they

A Replace all vaccines

B Chemically purify water

C Only infect humans

D Connect human, animal and environmental microbiomes, including AMR dynamics

ROUND 9

NEW TOPIC

AUDIENCE

Phages are part of ·One Health· thinking because they

A Replace all vaccines

B Chemically purify water

C Only infect humans

D Connect human, animal and environmental microbiomes, including AMR dynamics

10

BACTERIAL DEFENSES & PHAGE COUNTER-DEFENSE

Correct +10 | Wrong -5 | 30 sec

ROUND 10

NEW TOPIC

CRISPR-Cas adaptive immunity stores phage-derived sequences in the bacterial genome as

A Telomeres

B Spacers

C Operators

D Introns

ROUND 10

NEW TOPIC

CRISPR-Cas adaptive immunity stores phage-derived sequences in the bacterial genome as

A Telomeres

B Spacers

C Operators

D Introns

ROUND 10

NEW TOPIC

Anti-CRISPR (Acr) proteins protect phages by

A Blocking phage adsorption

B Degrading bacterial DNA

C Inducing host lysis early

D Inhibiting Cas proteins

ROUND 10

NEW TOPIC

Anti-CRISPR (Acr) proteins protect phages by

A Blocking phage adsorption

B Degrading bacterial DNA

C Inducing host lysis early

D Inhibiting Cas proteins

ROUND 10

NEW TOPIC

Restriction–modification systems defend bacteria by

A Pumping phages out by efflux

B Methylating phage tail fibers

C Cleaving unmethylated foreign DNA

D Blocking bacterial ribosomes

ROUND 10

NEW TOPIC

Restriction·modification systems defend bacteria by

A Pumping phages out by efflux

B Methylating phage tail fibers

C Cleaving unmethylated foreign DNA

D Blocking bacterial ribosomes

ROUND 10

NEW TOPIC

Abortive infection (Abi) systems protect a bacterial population by

A Repairing damaged DNA

B Triggering death or dormancy of the infected cell

C Preventing phage adsorption

D Degrading extracellular phages

ROUND 10

NEW TOPIC

Abortive infection (Abi) systems protect a bacterial population by

A Repairing damaged DNA

B Triggering death or dormancy of the infected cell

C Preventing phage adsorption

D Degrading extracellular phages

ROUND 10

NEW TOPIC

The arbitrium system lets temperate phages

A Repair their DNA after UV

B Sense a peptide cue of host density to choose lysis vs lysogeny

C Adsorb faster to pili

D Resist CRISPR targeting

ROUND 10

NEW TOPIC

The arbitrium system lets temperate phages

A Repair their DNA after UV

B Sense a peptide cue of host density to choose lysis vs lysogeny

C Adsorb faster to pili

D Resist CRISPR targeting

ROUND 10

NEW TOPIC

Phage VP882 eavesdrops on which bacterial signal to time its lysis decision?

A Teichoic acid

B Lipopolysaccharide

C A quorum-sensing autoinducer

D Host ATP levels

ROUND 10

NEW TOPIC

Phage VP882 eavesdrops on which bacterial signal to time its lysis decision?

A Teichoic acid

B Lipopolysaccharide

C A quorum-sensing autoinducer

D Host ATP levels

ROUND 10

NEW TOPIC

Phage resistance via loss of a surface receptor often costs the bacterium

A Reduced fitness or virulence, sometimes restored antibiotic sensitivity

B Prophage induction

C Immediate lysis

D CRISPR array loss

ROUND 10

NEW TOPIC

Phage resistance via loss of a surface receptor often costs the bacterium

A Reduced fitness or virulence, sometimes restored antibiotic sensitivity

B Prophage induction

C Immediate lysis

D CRISPR array loss

ROUND 10

NEW TOPIC

A classic phage counter-defense against restriction systems is

A DNA modification, e.g. glucosylated 5-hydroxymethylcytosine in T4

B Faster adsorption kinetics

C A longer tail

D A larger capsid

ROUND 10

NEW TOPIC

A classic phage counter-defense against restriction systems is

A DNA modification, e.g. glucosylated 5-hydroxymethylcytosine in T4

B Faster adsorption kinetics

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ROUND 10

NEW TOPIC

AUDIENCE

Superinfection exclusion means that

A. A resident prophage blocks secondary infection by related phages

B. Two phages co-infect one cell

C. Phages exclude antibiotics from the cell

D. Bacteria pump phages out by efflux

ROUND 10

NEW TOPIC

AUDIENCE

Superinfection exclusion means that

A. A resident prophage blocks secondary infection by related phages

B. Two phages co-infect one cell

C. Phages exclude antibiotics from the cell

D. Bacteria pump phages out by efflux

ROUND 10

NEW TOPIC

AUDIENCE

BREX, DISARM, Thoeris and Gabija are examples of

A Plasmid incompatibility groups

B Staining methods

C Phage families

D Bacterial anti-phage defense systems

ROUND 10

NEW TOPIC

AUDIENCE

BREX, DISARM, Thoeris and Gabija are examples of

A Plasmid incompatibility groups

B Staining methods

C Phage families

D Bacterial anti-phage defense systems

Break Time!

5 MIN

11

RAPID FIRE

Correct +5 | Wrong 0 | 60 sec

ROUND 11 - RAPID FIRE

Lytic Legends - question 1 of 8

Enzyme that degrades the peptidoglycan at the end of the lytic cycle

ROUND 11 - RAPID FIRE

Lytic Legends - question 1 of 8

Enzyme that degrades the peptidoglycan at the end of the lytic cycle

Endolysin (lysozyme)

ROUND 11 - RAPID FIRE

Lytic Legends - question 2 of 8

Term for a phage genome integrated in the host chromosome

ROUND 11 - RAPID FIRE

Lytic Legends - question 2 of 8

Term for a phage genome integrated in the host chromosome

Prophage

ROUND 11 - RAPID FIRE

Lytic Legends - question 3 of 8

UV light or mitomycin C triggers this event in a lysogen

ROUND 11 - RAPID FIRE

Lytic Legends - question 3 of 8

UV light or mitomycin C triggers this event in a lysogen

Prophage induction

ROUND 11 - RAPID FIRE

Lytic Legends - question 4 of 8

The protein shell that encloses the phage genome



ROUND 11 - RAPID FIRE

Lytic Legends - question 4 of 8

The protein shell that encloses the phage genome

Capsid

ROUND 11 - RAPID FIRE

Lytic Legends - question 5 of 8

The myovirus structure that contracts during infection



ROUND 11 - RAPID FIRE

Lytic Legends - question 5 of 8

The myovirus structure that contracts during infection

Tail sheath

ROUND 11 - RAPID FIRE

Lytic Legends - question 6 of 8

Number of progeny released per infected cell

ROUND 11 - RAPID FIRE

Lytic Legends - question 6 of 8

Number of progeny released per infected cell

Burst size

ROUND 11 - RAPID FIRE

Lytic Legends - question 7 of 8

Filamentous phage widely used for phage display

ROUND 11 - RAPID FIRE

Lytic Legends - question 7 of 8

Filamentous phage widely used for phage display

M13 (f1/fd)

ROUND 11 - RAPID FIRE

Lytic Legends - question 8 of 8

Ions added to media because many phages need them for adsorption

ROUND 11 - RAPID FIRE

Lytic Legends - question 8 of 8

Ions added to media because many phages need them for adsorption

Divalent cations (Ca2+ / Mg2+)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 1 of 8

Matrix component degraded by phage depolymerases in a biofilm



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 1 of 8

Matrix component degraded by phage depolymerases in a biofilm

Exopolysaccharide (EPS / capsule)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 2 of 8

Phage-mediated transfer of bacterial DNA between cells



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 2 of 8

Phage-mediated transfer of bacterial DNA between cells

Transduction

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 3 of 8

Microscopy that images phages in a near-native frozen-hydrated state

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 3 of 8

Microscopy that images phages in a near-native frozen-hydrated state

Cryo-EM (cryo-TEM)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 4 of 8

The phage life cycle that ends in host lysis



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 4 of 8

The phage life cycle that ends in host lysis

Lytic cycle

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 5 of 8

Extrachromosomal bacterial element that often carries AMR genes

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 5 of 8

Extrachromosomal bacterial element that often carries AMR genes

Plasmid

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 6 of 8

A phage particle that lacks its genome



ROUND 11 - RAPID FIRE

Biofilm Breakers - question 6 of 8

A phage particle that lacks its genome

Empty capsid (ghost)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 7 of 8

The workbench host bacterium of the T-phages

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 7 of 8

The workbench host bacterium of the T-phages

Escherichia coli

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 8 of 8

Ecological model: phages lyse the most abundant host (3 words, ·kill the ··)

ROUND 11 - RAPID FIRE

Biofilm Breakers - question 8 of 8

Ecological model: phages lyse the most abundant host (3 words, ·kill the ··)

Kill-the-winner

ROUND 11 - RAPID FIRE

Phage Pharmers - question 1 of 8

US regulatory route for compassionate phage use (3 letters + IND)



ROUND 11 - RAPID FIRE

Phage Pharmers - question 1 of 8

US regulatory route for compassionate phage use (3 letters + IND)

eIND (emergency/expanded-access IND)

ROUND 11 - RAPID FIRE

Phage Pharmers - question 2 of 8

Full name of PEG, used with NaCl to precipitate phages

ROUND 11 - RAPID FIRE

Phage Pharmers - question 2 of 8

Full name of PEG, used with NaCl to precipitate phages

Polyethylene glycol

ROUND 11 - RAPID FIRE

Phage Pharmers - question 3 of 8

Instrument that separates phages by buoyant density



ROUND 11 - RAPID FIRE

Phage Pharmers - question 3 of 8

Instrument that separates phages by buoyant density

(Ultra)centrifuge

ROUND 11 - RAPID FIRE

Phage Pharmers - question 4 of 8

Enzymes that cut DNA at specific sites, first studied in phage biology



ROUND 11 - RAPID FIRE

Phage Pharmers - question 4 of 8

Enzymes that cut DNA at specific sites, first studied in phage biology

Restriction endonucleases

ROUND 11 - RAPID FIRE

Phage Pharmers - question 5 of 8

Coat proteins used to display peptides on M13 (two names)

ROUND 11 - RAPID FIRE

Phage Pharmers - question 5 of 8

Coat proteins used to display peptides on M13 (two names)

pIII / pVIII

ROUND 11 - RAPID FIRE

Phage Pharmers - question 6 of 8

Route used to deliver phages to CF lungs in the 2025 first-in-human randomized trial

ROUND 11 - RAPID FIRE

Phage Pharmers - question 6 of 8

Route used to deliver phages to CF lungs in the 2025 first-in-human randomized trial

Nebulized / inhaled

ROUND 11 - RAPID FIRE

Phage Pharmers - question 7 of 8

Chelator of divalent cations that can block phage adsorption



ROUND 11 - RAPID FIRE

Phage Pharmers - question 7 of 8

Chelator of divalent cations that can block phage adsorption

EDTA

ROUND 11 - RAPID FIRE

Phage Pharmers - question 8 of 8

Phages that kill the host and never lysogenize are called

ROUND 11 - RAPID FIRE

Phage Pharmers - question 8 of 8

Phages that kill the host and never lysogenize are called

Virulent (strictly lytic) phages

ROUND 11 - RAPID FIRE

Virion Architects - question 1 of 8

Tubulin-like phage protein that forms the spindle of jumbo phages



ROUND 11 - RAPID FIRE

Virion Architects - question 1 of 8

Tubulin-like phage protein that forms the spindle of jumbo phages

PhuZ

ROUND 11 - RAPID FIRE

Virion Architects - question 2 of 8

Protein that builds the shell of the ·KZ ·phage nucleus·

ROUND 11 - RAPID FIRE

Virion Architects - question 2 of 8

Protein that builds the shell of the ·KZ ·phage nucleus·

Chimallin (ChmA)

ROUND 11 - RAPID FIRE

Virion Architects - question 3 of 8

Capsid vertex through which DNA enters during assembly



ROUND 11 - RAPID FIRE

Virion Architects - question 3 of 8

Capsid vertex through which DNA enters during assembly

Portal vertex (portal protein)

ROUND 11 - RAPID FIRE

Virion Architects - question 4 of 8

ATP-driven enzyme that packs DNA into the procapsid



ROUND 11 - RAPID FIRE

Virion Architects - question 4 of 8

ATP-driven enzyme that packs DNA into the procapsid

Terminase (large subunit)

ROUND 11 - RAPID FIRE

Virion Architects - question 5 of 8

Morphotype with a long, non-contractile tail



ROUND 11 - RAPID FIRE

Virion Architects - question 5 of 8

Morphotype with a long, non-contractile tail

Siphovirus

ROUND 11 - RAPID FIRE

Virion Architects - question 6 of 8

Geometric term for a capsid with 20 triangular faces



ROUND 11 - RAPID FIRE

Virion Architects - question 6 of 8

Geometric term for a capsid with 20 triangular faces

Icosahedral

ROUND 11 - RAPID FIRE

Virion Architects - question 7 of 8

Hollow tube through which the genome crosses the envelope in tailed phages



ROUND 11 - RAPID FIRE

Virion Architects - question 7 of 8

Hollow tube through which the genome crosses the envelope in tailed phages

Tail tube

ROUND 11 - RAPID FIRE

Virion Architects - question 8 of 8

The extra lipid layer present in ·6 but absent in T4



ROUND 11 - RAPID FIRE

Virion Architects - question 8 of 8

The extra lipid layer present in ϕ6 but absent in T4

Envelope (membrane)

12

TIE-BREAKER ROUND

Correct +20 | Wrong -10 | 60 sec | passed +5

ROUND 12

The T4 genome is circularly permuted and terminally redundant. Which packaging strategy generates this architecture?

A Headful packaging from concatemers

B cos-site cleavage of concatemers

C Protein-primed replication

D Rolling-circle of monomers

ROUND 12

The T4 genome is circularly permuted and terminally redundant. Which packaging strategy generates this architecture?

A Headful packaging from concatemers

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D Rolling-circle of monomers

ROUND 12

DNA replication of phage ·29 is primed by

A Host DnaA

B A primase (DnaG)

C A terminal protein

D A ribozyme

ROUND 12

DNA replication of phage ϕ29 is primed by

A Host DnaA

B A primase (DnaG)

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ROUND 12

In the lambda decision circuit, high MOI together with a poor physiological host state favours

A Pseudolysogeny

B Lysis

C Superinfection

D Lysogeny (stabilized cII)

ROUND 12

In the lambda decision circuit, high MOI together with a poor physiological host state favours

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ROUND 12

T4 glucosylates its 5-hydroxymethylcytosine DNA mainly to

A Protect the genome from host restriction endonucleases

B Aid adsorption

C Assist capsid assembly

D Speed up replication

ROUND 12

T4 glucosylates its 5-hydroxymethylcytosine DNA mainly to

A Protect the genome from host restriction endonucleases

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C Assist capsid assembly

D Speed up replication

ROUND 12

Which anti-phage system signals through a cyclic oligonucleotide (cGAMP) to activate a nuclease effector?

A Retrons

B Septu

C Gabija

D CBASS

ROUND 12

Which anti-phage system signals through a cyclic oligonucleotide (cGAMP) to activate a nuclease effector?

A Retrons

B Septu

C Gabija

D CBASS

ROUND 12

The lambda Red recombineering toolkit consists of the proteins

A EcoRI, EcoRV, ligase

B RecA, RecB, RecC

C Exo, Beta, Gam

D Int, Xis, Gam

ROUND 12

The lambda Red recombineering toolkit consists of the proteins

A EcoRI, EcoRV, ligase

B RecA, RecB, RecC

C Exo, Beta, Gam

D Int, Xis, Gam

ROUND 12

In the 2025 first-in-human randomized trial (BX004-A), the phage cocktail reached the CF lung by

A Intravenous infusion

B Nebulization

C Topical gel

D Oral capsules

ROUND 12

In the 2025 first-in-human randomized trial (BX004-A), the phage cocktail reached the CF lung by

A Intravenous infusion

B Nebulization

C Topical gel

D Oral capsules

ROUND 12

After a strong phage-driven selective sweep in a bacterial community, which metric is expected to DECREASE?

A Phage titer

B Shannon diversity index

C Burst size

D MOI

ROUND 12

After a strong phage-driven selective sweep in a bacterial community, which metric is expected to DECREASE?

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13

LYTIC LAUGHS

Correct +10 | Wrong 0 | 30 sec

ROUND 13

Why did the phage break up with the pilus?

A ·No chemistry in the periplasm.·

B ·It was too contractile.·

C ·Long-distance lysis never works.·

D ·The relationship was only surface-level attachment.·

ROUND 13

Why did the phage break up with the pilus?

A ·No chemistry in the periplasm.·

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D ·The relationship was only surface-level attachment.·

ROUND 13

What is a bacteriophage's favourite genre of music?

A Ambient · like the eclipse period

B Smooth jazz for slow adsorption

C Opera, because it's always dramatic lysis

D Lytic metal · full of breakdowns

ROUND 13

What is a bacteriophage's favourite genre of music?

A Ambient · like the eclipse period

B Smooth jazz for slow adsorption

C Opera, because it's always dramatic lysis

D Lytic metal · full of breakdowns

ROUND 13

Why don't phages use dating apps?

A No signal inside the gut

B They only ever swipe right on one specific host

C Too many matches cause lysis from without

D Their profiles are only ~50 kb

ROUND 13

Why don't phages use dating apps?

A No signal inside the gut

B They only ever swipe right on one specific host

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ROUND 13

Where does a phage see itself in 30 minutes?

A ·In a preprint.·

B ·At a plaquelytics retreat.·

C ·As ~200 copies of myself inside one very surprised bacterium.·

D ·Integrated, settled and dormant.·

ROUND 13

Where does a phage see itself in 30 minutes?

A ·In a preprint.·

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D ·Integrated, settled and dormant.·

ROUND 13

Why was the temperate phage terrible at commitment?

A It kept integrating into relationships and going dormant

B It lysed every relationship too fast

C It had no receptors for feelings

D It was too virulent to care

ROUND 13

Why was the temperate phage terrible at commitment?

A It kept integrating into relationships and going dormant

B It lysed every relationship too fast

C It had no receptors for feelings

D It was too virulent to care

ROUND 13

What did the depolymerase say arriving at the biofilm party?

A ·I brought capsules for the road.·

B ·Relax · I'll break down the matrix and free everyone.·

C ·More EPS for everyone!·

D ·Who invited the antibiotics?·

ROUND 13

What did the depolymerase say arriving at the biofilm party?

A ·I brought capsules for the road.·

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C ·More EPS for everyone!·

D ·Who invited the antibiotics?·

ROUND 13

Why do phages make bad stand-up comedians?

A Their delivery is always lytic · they kill the room

B They are too small for the microphone

C They only infect the hecklers

D They have no tail for timing

ROUND 13

Why do phages make bad stand-up comedians?

A Their delivery is always lytic · they kill the room

B They are too small for the microphone

C They only infect the hecklers

D They have no tail for timing

ROUND 13

What is a jumbo phage's excuse at the gym?

A ·I skip tail day.·

B ·My sheath is already too tight.·

C ·I carry a 500-kb genome · that's enough baggage.·

D ·No capsid space left.·

ROUND 13

What is a jumbo phage's excuse at the gym?

A ·I skip tail day.·

B ·My sheath is already too tight.·

C ·I carry a 500-kb genome · that's enough baggage.·

D ·No capsid space left.·

Thank you for your answers!

See the scoreboard for final standings